

SURAJ MERUVA

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Education

IIIT Naya Raipur B.Tech in Data Science & AI CGPA: 9.01
Aakash Institute Class XII – 94.5% JEE: 98th Percentile
Visakha Valley School Class X – 90%

Professional Experience

Summer Research Intern – IIT Kharagpur May 2025 – Jul 2025

- Worked on ECG/PPG biomedical signal processing; built deep learning pipelines for multiclass arrhythmia classification (NSR, AF, PAC/PVC) from smartwatch PPG using the Pulsewatch dataset.

Research Intern – NIT Warangal Dec 2024 – Feb 2025

- Researched 3D medical image segmentation; adapted prototype-based frameworks for thyroid nodule detection with uncertainty-aware training strategies.

Founder – Orbis Systems 2024 – Present

- Founded AI/ML startup; led end-to-end product development, system design, and client delivery across multiple intelligent software solutions.

AI/ML Engineer – Dept. of Forest Officer, Chhattisgarh Govt. Sep 2025 – Present

- Built Hindi voice-based Zoo AI logging system using Deepgram ASR and Gemini LLM; developed RBAC modules for inventory, medication, and analytics.

Full-Stack Engineer – Godavari Tourism

- Built booking platform with React, TypeScript, Firebase; implemented real-time Firestore sync and role-based admin dashboard.

Research & Patents

FileTract: AI-Powered Multi-File OCR *Patent Approval Pending*

- Confidence-weighted OCR pipeline for degraded documents with adaptive re-OCR and LLM-based field extraction.

Cross-Attention and AU-Guided Multimodal Hybrid Framework for Stress Detection *Journal Submitted*

- Multimodal fusion of facial action units and cross-attention; outperformed unimodal baselines on standard stress benchmarks.

Uncertainty-Aware Multimodal Deep Learning for Stress Detection via Audio-Visual Signals *ICAIHC 2026 – Submitted*

- Uncertainty-aware audio-visual fusion with epistemic modeling for robust stress prediction under noisy conditions.

Key Projects

CSC Sahayak – AI Co-Pilot for Common Service Centres *GSSoC Selected — 15+ Contributors — 30+ PRs*

- Built a 4-component AI ecosystem: Chrome Extension, FileTract Flask backend, React Desktop App, and Monitoring Dashboard for CSC operators across India.
- Bilingual (Hindi/English) AI assistant with Groq LLM; one-click form auto-fill via content scripts on government portals; OCR + LLM extraction (Gemini, Groq, Ollama) using a patent-eligible confidence-weighted pipeline.

DocuChat – Multimodal RAG Chatbot

- RAG chatbot for PDFs, images, audio, and web links with hybrid retrieval (FAISS + BM25) and FastAPI/Gemini backend.

Automated Satellite Encroachment Detection – CSIDC

- End-to-end pipeline using Playwright scraping, color-graded satellite image superimposition, and LLM visual reasoning for automated compliance detection and report generation.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript **Frontend:** React.js, Next.js, Tailwind CSS **Backend:** FastAPI, Node.js, Express.js

Databases: PostgreSQL, MongoDB, MySQL, Redis, Firebase Firestore **Vector Stores:** FAISS, ChromaDB, Elasticsearch

AI / ML: LLMs (Gemini, Groq), RAG, Hybrid Search, QLoRA, PEFT, Multimodal AI, Biomedical Signal Processing (ECG/PPG), Time-Series Deep Learning (BiLSTM, GRU), Computer Vision, 3D Medical Image Segmentation, OCR, XAI (SHAP, LIME, Grad-CAM), ASR

Tools: Git, Docker, Firebase, LangChain, Streamlit, Ollama, Postman

Awards & Achievements

- **Kaggle Notebooks Expert & Datasets Expert**
- 3rd Runner-Up, Technovate 6.0 Hackathon (Multimodal RAG Domain)
- 4th Place, Bit2Byte Hackathon

Certifications

- AI Engineer Agentic Track: The Complete Agent & MCP Course – Ed Donner (Udemy)
- AI Engineer Production Track: Deploy LLMs & Agents at Scale – Ed Donner (Udemy)
- AI Engineer Core Track (LLMs, RAG, QLoRA, Agents) – Ed Donner (Udemy)
- The Complete Full-Stack Web Development Bootcamp – Dr. Angela Yu (Udemy)
- Machine Learning A–Z + ChatGPT Prize [2025] – SuperDataScience
- Beginning C++ Programming – Tim Buchalka (Udemy)
- Android 14 & Kotlin Development Masterclass – Denis Panjuta (Udemy)